

Review retrospective:
the quartic–sextic odd-moment vanishing — closing the 2D lift family

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Abstract

A soundness review of `Laplace/TwoD/QuarticSexticOddMoments.lean`: the two parity-completion theorems stating that a mixed quartic–sextic 2D moment vanishes when at least one exponent is odd. Result: **a clean fidelity pass with no edits**. With this file, the entire 2D separable-lift family — the `AddSeparable` root, the generic `kthkth` trio and odd companion, and the `(2,3) QuarticSextic*` specialisations and cores — is reviewed end-to-end.

1 Setting

The `(2,3)` analogue of the generic `KthKthOddMoments`, and the odd-moment companion to the `QuarticSextic` core. It is the last unreviewed file of the `TwoD/` lift family.

2 What was reviewed

Two theorems: `gibbsExpectation_quarticSextic_odd_pow_pow_eq_zero` ($\langle x^{2j+1}y^n \rangle = 0$ for any n) and `gibbsExpectation_quarticSextic_pow_odd_pow_eq_zero` ($\langle x^m y^{2k+1} \rangle = 0$ for any m). Each factorises via the core’s `gibbsExpectation_quarticSextic_factor` and kills the odd 1D factor with `quartic_expected_value_odd` (resp. `sextic_expected_value_odd`); the product is zero by `zero_mul/mul_zero`.

3 Fidelity / soundness findings

Sound; no mismatch. As in the generic companion, the non-odd factor needs only integrability — supplied by `quartic_integrable_pow_pot/sextic_integrable_pow_pot` for every natural exponent — so each theorem holds for arbitrary n (resp. m), and the two cover every “at least one odd” case including odd–odd. The docstrings are accurate.

4 Simplifications applied

None — the bucket was empty. The file has no open statements, its single import is load-bearing, and the closing `simp` does real work (`zero_mul/mul_zero`). The reviewer’s only suggestion — merging the `rw` and `simp` into one `simp` — is cosmetic proof-golf, declined. No worktree was created.

5 Disagreements and open questions

None.

6 Lessons

- **The 2D lift family is internally consistent end-to-end.** Reviewing the root, the generic trio+odd companion, and the $(2,3)$ specialisations and cores in dependency order surfaced exactly one genuine fidelity drift (the partition-asymptotic doc bullet naming the wrong theorem), a scatter of dead `opens` (Real always, MeasureTheory only in the def-stated files), two ill-typed/mislabelled docstrings, and three clean empty-bucket passes — and no mathematical error anywhere. The closed-form constants all trace back to the 1D `quartic_partition/sextic_partition` and `*_expected_value_*` lemmas.

7 Follow-ups

- The `TwoD/` lift family is now fully reviewed. Remaining laplace surface: the larger `OneD/` files not yet covered (`NearDegenerate`, `UniversalAsymptotics`, `HarmonicCovGlobalConstancy`, `HarmonicPerturbedMoments`, the `Harmonic*` monomials) and the `Multi/Covariance*` trio (sectioned).
- The accumulated forward tides: `AddSeparable`'s direct-Mathlib product-integral import; the batched `TwoD/ t>0` module-docstring hygiene; the `simpa/single-use-have` proof-golf the reviewer flagged across the family.